CS383 · SOFTWARE ENGINEERING

Lecture 2 Introduction (II)

Midterm summary

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01 The four activities: Process activities

Every software building process has four fundamental activities:

#	Activity	What happens in it
1	**Software specification**	We define what software will be built and the constraints on its operation.
2	**Software development**	We design the software and implement it according to the specification.
3	**Software verification and validation**	We make sure the software is what the customer needs.
4	**Software evolution**	We modify the software so it keeps up with changes in the customer and the market.

Appeared in the midterm

A question asked you to write the four activities in their logical order (2 marks).

02 Software engineering and the Web

In the last twenty years, systems have come to be built on the web and opened from the browser. So the software does not need to be installed on the device, and anyone in the

world can reach it right away.

What did the developer gain?

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Faster software delivery
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Cheaper to change and upgrade
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Reduce cost
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What changed in the way of developing?

- **Software reuse** became essential and important.
- Specifying all the requirements in advance is no longer practical, so we use **Incremental development**.
- The drawback: the user interface is limited by the capabilities of the browser.

03 Software as a Service (SaaS)

Instead of installing the whole program on the device, we take it as a service. The program runs on a cloud and we reach it through the web.

Term	What it provides
Platform	Cloud components and a framework for building programs. It frees you from worrying about the operating system and storage.
Infrastructure	Scalable resources: servers, network, storage, provided through virtualization.

Benefits	Drawbacks
Pay as you go You pay for what you use, monthly or yearly.	**Lack of control** You have little control over it.
Accessible anywhere You can reach it from any place.	**Vulnerable: security and trust** It is exposed to security and trust problems.
No need for expensive computers A cheap device is enough.	**Limited range of applications** Only a few applications are available.
	Internet dependent It does not work without the internet.
	Performance issues It can be slow.

Appeared in the midterm

An MCQ question about one of the drawbacks of SaaS. The correct answer was:

> Security/trust vulnerabilities and internet dependence

04 SE Ethics: the eight principles

The **IEEE/ACM Code of Ethics** has eight principles. In the exam you are given a description and you write the name of the principle.

Source: the text of the Code itself; the lecture file has no ethics slides.

Principle	Meaning	Words that point to it
PUBLIC	Act in a way that is consistent with the public interest.	responsibility for their own work
CLIENT AND EMPLOYER	Serve the interest of the client and the employer, without going against the public interest.	confidential information
PRODUCT	The product and its modifications meet the highest professional standards.	appropriate method, testing, integrity of data
JUDGMENT	Integrity and independence in professional judgment.	conflicts of interest
MANAGEMENT	Managers support the ethical management of development and maintenance.	informed of standards
PROFESSION	Raise the integrity and reputation of the profession.	reputation of the profession
COLLEAGUES	Be fair to your colleagues and support them.	fair hearing, colleague
SELF	Keep learning throughout your life and follow the ethics personally.	improve their knowledge, prejudices

05 How does it appear in the midterm?

The ethics question appeared in three midterms: 462 for 8 marks, 451 for 4 and 461 for 2.

Here are the items of midterm 462 and their answers:

Description	Principle
Give a fair hearing to the opinions, concerns, or complaints of a colleague.	**COLLEAGUES**
Improve their knowledge of this Code, its interpretation, and its application to their work.	**SELF**
Ensure that software engineers are informed of standards before being held to them.	**MANAGEMENT**
Ensure an appropriate method is used for any project on which they work or propose to work.	**PRODUCT**
Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.	**JUDGMENT**
Maintain the integrity of data, being sensitive to outdated or flawed occurrences.	**PRODUCT**
Accept full responsibility for their own work.	**PUBLIC**
Not give unfair treatment to anyone because of any irrelevant prejudices.	**SELF**

Watch out for these two

Conflicts of interest is answered with **JUDGMENT**, and integrity of data is answered with **PRODUCT**.

06 Common mistakes

Mistake	Correct
Putting conflicts of interest under Management	It is under **JUDGMENT**.
Putting integrity of data under Profession	It is under **PRODUCT**.
Saying SaaS gives you full control	The opposite, **Lack of control** is one of its drawbacks.
Reversing the order of the four activities	Specification, then Development, then V&V, then Evolution.

07 Test yourself

1. List two benefits and two drawbacks of SaaS.
2. In which activity do we make sure the software is what the customer needs?
3. Which principle covers: **Keep private any confidential information**?
4. Which principle covers: **Ensure adequate testing, debugging, and review of software**?
5. Which principle covers: **Assist colleagues in professional development**?

Answers

#	Answer
1	Two benefits: Pay as you go, Accessible anywhere. Two drawbacks: Lack of control, Internet dependent.
2	**Software verification and validation**.
3	**CLIENT AND EMPLOYER**.
4	**PRODUCT**.
5	**COLLEAGUES**.